

Supplemental Material:
Sign structure of thermal Hall conductivity and topological magnons
for in-plane field polarized Kitaev magnets

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S1. DETAILS OF ANALYTICAL PROOFS

Here we provide more details to the analytical proofs of the two theorems concerning the sign structure of the magnon thermal Hall conductivity κ_{xy} in the $K\Gamma\Gamma'$ model, which are the main results of this study.

Theorem 1. κ_{xy} in the polarized states under magnetic fields along the a and $-a$ directions differ by a minus sign.

Theorem 2. κ_{xy} in the polarized state under a magnetic field along the b direction is zero.

Preliminaries. The $K\Gamma\Gamma'$ model under a magnetic field is given in the main text. To derive the linear spin wave Hamiltonian of a polarized state, the coordinate frames of all spins are first rotated uniformly such that the z -axes align with the spins, $\mathbf{S}_i = R\tilde{\mathbf{S}}_i$. In the rotated frame, the Hamiltonian is $H = \sum_{\lambda \in \{x,y,z\}} \sum_{\langle ij \rangle \in \Lambda} \tilde{\mathbf{S}}_i^T \tilde{H}_\lambda \tilde{\mathbf{S}}_j - \sum_i \tilde{\mathbf{h}} \cdot \tilde{\mathbf{S}}_i$, where $\tilde{H}_\lambda = R^T H_\lambda R$ and $\tilde{\mathbf{h}} = R^T \mathbf{h}$. We choose the rotation matrix $R \in SO(3)$ according to (1). For convenience of latter discussion, we introduce the notation

$$\tilde{H}_\lambda = \begin{pmatrix} \tilde{h}_{11}^\lambda & \tilde{h}_{12}^\lambda & \tilde{h}_{13}^\lambda \\ \tilde{h}_{21}^\lambda & \tilde{h}_{22}^\lambda & \tilde{h}_{23}^\lambda \\ \tilde{h}_{31}^\lambda & \tilde{h}_{32}^\lambda & \tilde{h}_{33}^\lambda \end{pmatrix} \quad (\text{S1})$$

and notice that $\tilde{H}_\lambda^T = \tilde{H}_\lambda$. Then, we apply the Holstein Primakoff transformation,

$$\tilde{S}_i^z = S - b_i^\dagger b_i = S - n_i, \quad (\text{S2a})$$

$$\tilde{S}_i^x = \frac{\sqrt{2S - n_i} b_i + b_i^\dagger \sqrt{2S - n_i}}{2}, \quad (\text{S2b})$$

$$\tilde{S}_i^y = \frac{\sqrt{2S - n_i} b_i - b_i^\dagger \sqrt{2S - n_i}}{2i}, \quad (\text{S2c})$$

and keep only terms up to second order in the bosonic operators b and $b^\dagger$. A subsequent Fourier transformation yields the linear spin wave Hamiltonian in momentum space, $H/S = \sum_{\mathbf{k}} \Psi_{\mathbf{k}}^\dagger \mathbf{D}_{\mathbf{k}} \Psi_{\mathbf{k}}$, where $\mathbf{D}_{\mathbf{k}}$ is a four dimensional Hermitian matrix and $\Psi_{\mathbf{k}} = (b_{1\mathbf{k}}, b_{2\mathbf{k}}, b_{1-\mathbf{k}}^\dagger, b_{2-\mathbf{k}}^\dagger)$. $\mathbf{D}_{\mathbf{k}}$ assumes the form

$$\mathbf{D}_{\mathbf{k}} = \begin{pmatrix} \mathbf{A}_{\mathbf{k}} & \mathbf{B}_{\mathbf{k}} \\ \mathbf{B}_{-\mathbf{k}}^* & \mathbf{A}_{-\mathbf{k}}^T \end{pmatrix}, \quad (\text{S3a})$$

$$\mathbf{A}_{\mathbf{k}} = \begin{pmatrix} h & 0 \\ 0 & h \end{pmatrix} + \sum_{\lambda} \begin{pmatrix} -\tilde{h}_{33}^\lambda & \frac{1}{2}(\tilde{h}_{11}^\lambda + \tilde{h}_{22}^\lambda) e^{i\mathbf{k} \cdot \delta_\lambda} \\ \frac{1}{2}(\tilde{h}_{11}^\lambda + \tilde{h}_{22}^\lambda) e^{-i\mathbf{k} \cdot \delta_\lambda} & -\tilde{h}_{33}^\lambda \end{pmatrix}, \quad (\text{S3b})$$

$$\mathbf{B}_{\mathbf{k}} = \sum_{\lambda} \frac{1}{2} \left(\tilde{h}_{11}^\lambda + 2i\tilde{h}_{12}^\lambda - \tilde{h}_{22}^\lambda \right) \begin{pmatrix} 0 & e^{i\mathbf{k} \cdot \delta_\lambda} \\ e^{-i\mathbf{k} \cdot \delta_\lambda} & 0 \end{pmatrix}. \quad (\text{S3c})$$

Some remarks are in order. First, the rotated hamiltonian (S1) is real. Second, only the entries $\tilde{h}_{11}^\lambda$, $\tilde{h}_{12}^\lambda = \tilde{h}_{21}^\lambda$, $\tilde{h}_{22}^\lambda$, and $\tilde{h}_{33}^\lambda$ in (S1) are relevant to the linear spin wave Hamiltonian, as shown in (S3a)-(S3c). Third, only $\tilde{h}_{12}^\lambda$ and $\tilde{h}_{21}^\lambda$ are multiplied by the imaginary unit i in $\mathbf{D}_{\mathbf{k}}$, as shown in (S3a)-(S3c).

Finally, the reciprocal lattice vectors $\mathbf{b}_1 = (4\pi/\sqrt{3}a)[\cos(\pi/6)\hat{\mathbf{x}} - \sin(\pi/6)\hat{\mathbf{y}}]$ and $\mathbf{b}_2 = (4\pi/\sqrt{3}a)\hat{\mathbf{y}}$ are defined through the primitive lattice vectors $\mathbf{a}_1 = a\hat{\mathbf{x}}$ and $\mathbf{a}_2 = a[\cos(\pi/3)\hat{\mathbf{x}} + \sin(\pi/3)\hat{\mathbf{y}}]$, where $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ are unit vectors in the a and b directions respectively. The first Brillouin zone is equivalent to the following parallelogram in reciprocal space,

$$\mathbf{k} = k_1 \hat{\mathbf{b}}_1 + k_2 \hat{\mathbf{b}}_2, \quad k_1 \in [0, 2\pi), \quad k_2 \in [0, 2\pi); \quad \hat{\mathbf{b}}_1 \equiv \frac{2}{\sqrt{3}a} \left(\cos \frac{\pi}{6} \hat{\mathbf{x}} - \sin \frac{\pi}{6} \hat{\mathbf{y}} \right), \quad \hat{\mathbf{b}}_2 \equiv \frac{2}{\sqrt{3}a} \hat{\mathbf{y}}. \quad (\text{S4})$$

If we write $\mathbf{k} = k_x \hat{\mathbf{x}} + k_y \hat{\mathbf{y}}$, then (k_1, k_2) and (k_x, k_y) are related by a linear transformation,

$$\begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ \sin \frac{\pi}{6} & \cos \frac{\pi}{6} \end{pmatrix} \begin{pmatrix} k_x \\ k_y \end{pmatrix}. \quad (\text{S5})$$

Proof of Theorem 1. The rotation matrices (1) for the polarized states along the a and $-a$ directions are, respectively,

$$R^a = \begin{pmatrix} -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \end{pmatrix}, \quad (\text{S6a})$$

$$R^{\bar{a}} = \begin{pmatrix} -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{pmatrix}. \quad (\text{S6b})$$

The explicit forms of the rotated spin Hamiltonians (S1) of the a and $-a$ polarized states are, respectively,

$$\tilde{H}_x^a = \begin{pmatrix} \frac{K+2\Gamma+4\Gamma'}{3} & \frac{K-\Gamma+\Gamma'}{\sqrt{6}} & \frac{-K+\Gamma-\Gamma'}{3\sqrt{2}} \\ \frac{K-\Gamma+\Gamma'}{\sqrt{6}} & \frac{K-2\Gamma'}{2} & \frac{-K-2\Gamma+2\Gamma'}{2\sqrt{3}} \\ \frac{-K+\Gamma-\Gamma'}{3\sqrt{2}} & \frac{-K-2\Gamma+2\Gamma'}{2\sqrt{3}} & \frac{K-4\Gamma-2\Gamma'}{6} \end{pmatrix}, \quad \tilde{H}_y^a = \begin{pmatrix} \frac{K+2\Gamma+4\Gamma'}{3} & \frac{-K+\Gamma-\Gamma'}{\sqrt{6}} & \frac{-K+\Gamma-\Gamma'}{3\sqrt{2}} \\ \frac{-K+\Gamma-\Gamma'}{\sqrt{6}} & \frac{K-2\Gamma'}{2} & \frac{K+2\Gamma-2\Gamma'}{2\sqrt{3}} \\ \frac{-K+\Gamma-\Gamma'}{3\sqrt{2}} & \frac{K+2\Gamma-2\Gamma'}{2\sqrt{3}} & \frac{K-4\Gamma-2\Gamma'}{6} \end{pmatrix}, \quad \tilde{H}_z^a = \begin{pmatrix} \frac{K+2\Gamma+4\Gamma'}{3} & 0 & \frac{\sqrt{2}(K-\Gamma+\Gamma')}{3} \\ 0 & -\Gamma & 0 \\ \frac{\sqrt{2}(K-\Gamma+\Gamma')}{3} & 0 & \frac{2K+\Gamma-4\Gamma'}{3} \end{pmatrix}, \quad (\text{S7a})$$

$$\tilde{H}_x^{\bar{a}} = \begin{pmatrix} \frac{K+2\Gamma+4\Gamma'}{3} & \frac{-K+\Gamma-\Gamma'}{\sqrt{6}} & \frac{K-\Gamma+\Gamma'}{3\sqrt{2}} \\ \frac{-K+\Gamma-\Gamma'}{\sqrt{6}} & \frac{K-2\Gamma'}{2} & \frac{-K-2\Gamma+2\Gamma'}{2\sqrt{3}} \\ \frac{K-\Gamma+\Gamma'}{3\sqrt{2}} & \frac{-K-2\Gamma+2\Gamma'}{2\sqrt{3}} & \frac{K-4\Gamma-2\Gamma'}{6} \end{pmatrix}, \quad \tilde{H}_y^{\bar{a}} = \begin{pmatrix} \frac{K+2\Gamma+4\Gamma'}{3} & \frac{K-\Gamma+\Gamma'}{\sqrt{6}} & \frac{K-\Gamma+\Gamma'}{3\sqrt{2}} \\ \frac{K-\Gamma+\Gamma'}{\sqrt{6}} & \frac{K-2\Gamma'}{2} & \frac{K+2\Gamma-2\Gamma'}{2\sqrt{3}} \\ \frac{K-\Gamma+\Gamma'}{3\sqrt{2}} & \frac{K+2\Gamma-2\Gamma'}{2\sqrt{3}} & \frac{K-4\Gamma-2\Gamma'}{6} \end{pmatrix}, \quad \tilde{H}_z^{\bar{a}} = \begin{pmatrix} \frac{K+2\Gamma+4\Gamma'}{3} & 0 & \frac{\sqrt{2}(-K+\Gamma-\Gamma')}{3} \\ 0 & -\Gamma & 0 \\ \frac{\sqrt{2}(-K+\Gamma-\Gamma')}{3} & 0 & \frac{2K+\Gamma-4\Gamma'}{3} \end{pmatrix}, \quad (\text{S7b})$$

while the Zeeman term has the same form for any polarized state, regardless of the field direction. The most important feature in (S7a) and (S7b) is that, among the entries relevant to the linear spin wave Hamiltonian, $\tilde{h}_{12}^\lambda$ and $\tilde{h}_{21}^\lambda$ change sign, while $\tilde{h}_{11}^\lambda$, $\tilde{h}_{22}^\lambda$, and $\tilde{h}_{33}^\lambda$ remain invariant, when the field direction is flipped from a to $-a$. (S3a)-(S3c) then imply that the linear spin wave Hamiltonians of the a and $-a$ polarized states are related by $D_{\mathbf{k}}^{\bar{a}} = (D_{-\mathbf{k}}^a)^*$. The rest of the proof is contained in the main text.

Proof of Theorem 2. The rotation matrix (1) for the polarized state along the b direction is

$$R^b = \begin{pmatrix} 0 & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -1 & 0 & 0 \end{pmatrix}. \quad (\text{S8})$$

The explicit form of the rotated spin Hamiltonian (S1) of the b polarized state is

$$\tilde{H}_x^b = \begin{pmatrix} 0 & \frac{\Gamma+\Gamma'}{\sqrt{2}} & \frac{-\Gamma+\Gamma'}{\sqrt{2}} \\ \frac{\Gamma+\Gamma'}{\sqrt{2}} & \frac{K+2\Gamma'}{2} & \frac{K}{2} \\ \frac{-\Gamma+\Gamma'}{\sqrt{2}} & \frac{K}{2} & \frac{K-2\Gamma'}{2} \end{pmatrix}, \quad \tilde{H}_y^b = \begin{pmatrix} 0 & \frac{\Gamma+\Gamma'}{\sqrt{2}} & \frac{\Gamma-\Gamma'}{\sqrt{2}} \\ \frac{\Gamma+\Gamma'}{\sqrt{2}} & \frac{K+2\Gamma'}{2} & -\frac{K}{2} \\ \frac{\Gamma-\Gamma'}{\sqrt{2}} & -\frac{K}{2} & \frac{K-2\Gamma'}{2} \end{pmatrix}, \quad \tilde{H}_z^b = \begin{pmatrix} K & \sqrt{2}\Gamma' & 0 \\ \sqrt{2}\Gamma' & \Gamma & 0 \\ 0 & 0 & -\Gamma \end{pmatrix}. \quad (\text{S9})$$

The most important feature in (S9) is that, among the matrix elements relevant to the linear spin wave Hamiltonian, $\tilde{h}_{ij}^x = \tilde{h}_{ij}^y$. On the other hand, the displacements (between unit cells) along the x , y , and z bonds are $\delta_x = (0, -1)$, $\delta_y = (1, -1)$, and $\delta_z = (0, 0)$, as measured by the primitive lattice vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ (see Fig. 1). Therefore, the phase factors in (S3b) and (S3c) are momentum dependent only along the x and y bonds. They are e^{-ik_2} and $e^{i(k_1-k_2)}$ (and their complex conjugates). By (S9), these two phase factors share the same coefficient in the linear spin wave Hamiltonian (S3a) of the b polarized state. We may as well say that $D_{\mathbf{k}}^b$ depends on k_1 and k_2 only through the combination $e^{-ik_2} + e^{i(k_1-k_2)} = 2 \cos(k_x a/2) e^{-i\sqrt{3}k_y a/2}$ (and its complex conjugate). The rest of the proof is contained in the main text.

S2. COMPARISON TO INELASTIC NEUTRON SCATTERING EXPERIMENT

The inelastic neutron scattering (INS) experiment in Ref. [22] is performed under a magnetic field along the a direction. Therefore, we can compare the magnon dispersion in Fig. 3 of the main text to the INS spectra of Ref. [22] at relevant

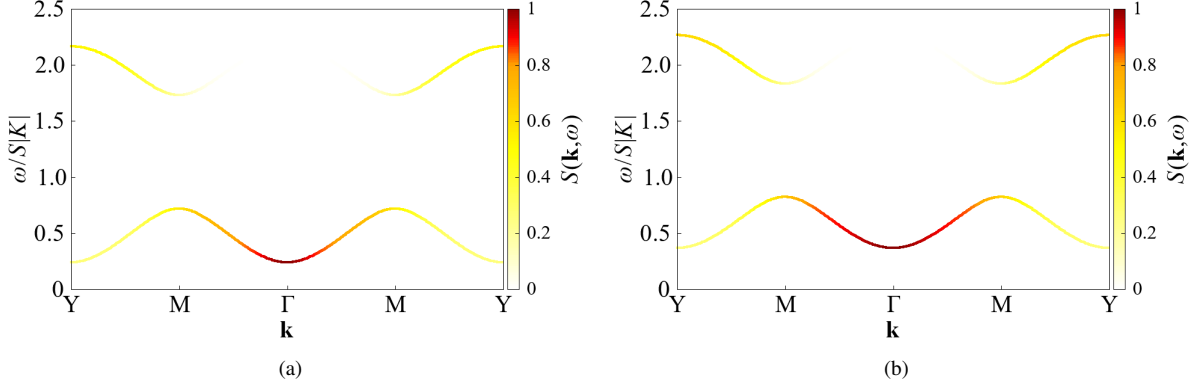


FIG. S1. Dynamical spin structure factors of the polarized states at the magnetic fields (a) $h/S|K| = 0.1$ and (b) 0.2 along the a direction, which approximately equal to 3 and 6 T, for $(K, \Gamma, \Gamma') = (-1, 0.2, -0.02)$. In each case, the intensity is normalized such that $\max S(\mathbf{k}, \omega) = 1$.

field strengths. As in the main text, we consider the $K\Gamma\Gamma'$ model with $(K, \Gamma, \Gamma') = (-1, 0.2, -0.02)$ under a magnetic field $\mathbf{h}$ along the a direction. For the convenience of comparison, we consider the field $\mathbf{B}$ defined by $g\mu_B\mathbf{B} \cdot \mathbf{S}_i \equiv \mathbf{h} \cdot \mathbf{S}_i$, which is measured in units of Tesla. For α - RuCl_3 , we may take $S = 1/2$, $|K| \approx 80 \text{ K} \approx 8 \text{ meV}$, and $g \approx 2.3$. Therefore, $h/S|K| = 0.1$ and 0.2 correspond to $B \approx 3 \text{ T}$ and 6 T respectively. The magnon gaps of the polarized states at these fields are $\Delta/S|K| = 0.238$ and 0.369 , which correspond to 1 and 1.5 meV respectively. Figs. 2c-2e in Ref. [22] suggest that the excitation gap varies between 1 and 1.5 meV when the field ranges from 4 to 8 T. Therefore, the magnon gap obtained from our calculation is numerically similar to the excitation gap obtained in the INS experiment. This result is encouraging, though we cannot make an absolutely precise numerical comparison because the magnitude of the Kitaev interaction and the g factor are approximate. Also, fine tuning the interaction parameters and magnetic field may lead to an even better agreement.

Within the framework of linear spin wave theory, we can further calculate the dynamical spin structure factor of the polarized state and compare it with the INS spectra at intermediate fields. The dynamical structure factor is given by

$$S(\mathbf{k}, \omega) = \int dt e^{i\omega t} \sum_{ij} \langle \mathbf{S}_i(t) \cdot \mathbf{S}_j(0) \rangle e^{i\mathbf{k} \cdot (\mathbf{r}_j - \mathbf{r}_i)}$$

Keeping only the inelastic contribution, we plot $S(\mathbf{k}, \omega)$ along the line of momenta Y-M- Γ -M-Y (which correspond to the labels $-1.0, -0.5, 0, 0.5, 1.0$ on the horizontal axes in Figs. 2a-2f of Ref. [22]) for the same set of interaction parameters, at the fields $h/S|K| = 0.1$ and 0.2 along the a direction. As shown in Figs. S1a and S1b, the intensity is the highest at the Γ point, which is same as in the INS spectra at intermediate fields. While our semiclassical analysis of the polarized state agrees well with Ref. [22] in terms of the excitation gap and the location of the strongest signal, it is unable to explain the excitation continuum seen in the INS experiment, which probably requires the consideration of large quantum effects.